Usability Testing Report: Software Engineering

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1 Revision History

Date	Developer	Notes
March 29th	Team #16, The Chill Guys	Initial outline and structure of the report based on the usability testing plan.
March 29th	Swesan Pathmanathan	Fill out details of how testing was carried out, and include the results of testing along with metric details.
April 5th	Hamzah Rawasia	Cleaned up the report, fixed formatting, and added the post-study interface revisions section.

2 Symbols, Abbreviations and Acronyms

symbol	description
HCI	Human–Computer Interaction
SUS	System Usability Scale
CSUQ	Computer System Usability Questionnaire (optional instrument)
UT	Usability test scenario; `UT-1`–`UT-7`; `UT-2R` is reference-only
FR	Functional Requirement (from SRS)
NFR	Nonfunctional Requirement (from SRS)
SRS	Software Requirements Specification
API	Application Programming Interface
T	Task identifier in session scripts (e.g., T1, T2)

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3 Introduction

3.1 Purpose and Scope

This report describes usability testing for Hitchly: what we wanted to learn, which tasks participants performed, what we asked them afterward, and how we measured success. It covers ten moderated sessions ($N = 10$). Task design follows the same functional areas as the Verification and Validation (VnV) report (`docs/VnVReport/VnVReport.tex`), so usability findings can be read next to functional test results.

The evaluation covers the mobile app (and related steps such as email verification) for McMaster riders and drivers. It focuses on learnability, efficiency, recovering from mistakes, and satisfaction, not on replacing automated or unit tests.

3.2 Relation to Requirements

Objectives also match the SRS themes for clear, low-friction flows. Where it helps, issues and recommendations can be tied to FR/NFR identifiers.

3.3 Document Structure

Section 4 lists goals. Section 5 covers participants, setting, and procedure. Section 6 lists task scenarios. Section 7 lists moderator prompts and surveys. Section 8 defines measures. Sections 9 and 10 give results and recommendations. Appendix 10.4 includes questionnaires and the session log template.

4 Goals of Usability Testing

These goals shape each session and the success criteria in Section 6. They map to the metrics in Section 8.

1. **Onboarding and verification:** See whether participants can register with a McMaster-affiliated account, complete verification, and choose a role, **including** understanding and fixing sign-up errors (wrong email format, weak password, missing driver license when required).

2. **Rider journey:** See whether they can find or request a ride, open trip details when needed, and **read the estimated cost** shown on match proposals.

3. **Driver journey:** See whether they can post a ride with direction and schedule, and work with passenger requests as the app presents them.

4. **Matching and trip state:** See whether they understand proposals, **accept or decline** matches or requests, handle "no match" or blocked states, and move between related screens without getting lost.

5. **After the ride:** See whether they can **submit a review** and **submit a report** through the in-app safety flow when the moderator provides a suitable trip state.

6. **Profile:** See whether they can **update profile information** (and optionally find settings) without undue friction.

7. **Error recovery:** Note hesitation, wrong taps, and dead ends, and whether participants recover with at most reasonable moderator help.

8. **Satisfaction:** Capture perceived ease and confidence with SUS and short debrief questions.

5 Methodology

5.1 Study Design

We conducted **moderated** sessions using a **think-aloud** approach. Participants worked through the task scenarios in Section 6 while a moderator observed and took notes. Section 9 summarizes outcomes for $N = 10$.

5.2 Participants

- **Sample:** Ten participants (P1–P10). Rider- vs. driver-heavy tasks followed the distribution in Section 9.
- **Role mix:** Six sessions emphasized **rider** tasks (UT-2); four emphasized **driver** tasks (UT-3). Everyone completed UT-1, UT-4, UT-6, UT-7, and UT-5 where the script applied; UT-6 and UT-7 required moderator-prepared trip state.
- **Scope: UT-2R** (recurring rides) was **not tested**; drivers used **one-time** rides only in UT-3.

5.3 Environment and Materials

- **Devices:** Two Android phones running the Hitchly Expo client on the same backend.
- **API and database: Render**-hosted API; **Neon** (PostgreSQL) for data.
- **Network:** Stable Wi-Fi; same `EXPO_PUBLIC_API_URL` on both devices (see `apps/mobile/lib/trpc.ts`).
- **Reset between participants:** Test accounts were removed and flows restarted from registration when needed so trips, requests, and matches did not carry over incorrectly.

5.3.1 Trip flows (configuration used)

Recurring rides (UT-2R): Not evaluated; see Section 6 for future reference.
Accounts: New `@mcmaster.ca` test accounts per reset when required; email verification as in UT-1; home address set before rider search or driver post when those tasks ran.

5.4 Session Outline

1. Briefing and consent (Appendix 10.4).
2. Optional background questionnaire.

3. Task order: UT-1 → UT-2 or UT-3 (by participant) → UT-4 → UT-6 → UT-7 → UT-5. Moderator may insert short setup steps (e.g. address) without changing task IDs.
4. After each task, prompts from Section 7 (including ease rating).
5. SUS after tasks (Appendix 10.4).
6. Debrief and session log update before reset or next participant.

5.5 Data Recording

For each task, record start and end time, success or failure, assists (light hint vs. direct help), and short notes (confusion, paraphrased comments). Use the session log fields in Appendix 10.4.

6 Task Scenarios

Each scenario supports the goals in Section 4 and traces to the functional evaluation areas in the Verification and Validation report (`docs/VnVReport`). Navigation details match the Expo app in `apps/mobile` as of this document. Success criteria are **binary** unless noted. **UT-2R** (recurring rides) was not part of this study; drivers used one-time rides only in UT-3.

6.1 Mapping to VnV functional areas

Table 1 links each task to the VnV report areas (admin-only checks, such as reports in an admin dashboard, are out of scope for end-user tasks).

6.2 Scenario UT-1: Sign-up, verification, and input errors (Area 1)

Part A — successful path

- **Context:** New McMaster student; Hitchly just installed.
- **Task:** Create an account with a McMaster-affiliated email, complete verification as required, and reach the main screen after sign-in.

Table 1: Task scenarios mapped to VnV functional evaluation areas.

VnV area	**Task ID(s)**	**Covered in usability testing**
1 – User verification	UT-1	Sign-up, email verification, and clear handling of invalid or incomplete inputs.
2 – User profile	UT-5	Updating profile fields; address and role as needed for other tasks.
3 – Ride matching	UT-2, UT-3, UT-4	Search and proposals; posting a ride; accept, decline, no match, or blocked states.
4 – Cost estimation	UT-2, UT-4	Reading the estimated share on a match card; optional moderator-prepared state if pricing updates when another rider joins.
5 – Review and reporting	UT-6, UT-7	Submitting a rating after a trip; submitting a report through the in-app safety flow.

- **Success:** Account created; verification step completed (or correctly attempted for the test account); participant reaches the post-auth home with at most one standard hint from the moderator.

Part B — invalid or incomplete inputs

- **Task:** With moderator guidance, try at least one invalid case: wrong email format, password below the app's rules, or (for driver test accounts) attempting driver registration without a required license upload if the flow asks for it. Correct inputs until errors clear or the participant explains the message.
- **Success:** The app shows understandable errors for bad input; the participant recovers without repeated moderator help beyond one hint. Judge clarity and recovery, not server-side test-case IDs.

6.3 Scenario UT-2: Rider — search, cost, pass, and request (Areas 3 and 4)

Prerequisite: Home address on profile. If missing, the app prompts to open profile; moderator confirms a seeded account or allows a short address sub-task.

- **Start:** Rider home; open find-a-ride if needed (e.g. from "My Rides").
- **Task:**
  1. Set route direction (to or from McMaster), date, and arrival time per moderator script.
  2. If travelling to campus, pick the moderator's drop-off from the campus list.
  3. Run search and wait until results load.
  4. On at least one match card, find and say aloud the **estimated cost** (dollar amount).
  5. If two or more cards appear: swipe left on one card to pass, then swipe right on another to request. If only one card: swipe left to pass, then use new search or note empty deck per script.

6. If there are no matches, read the empty state (e.g. no matches found) and optionally start a new search.
7. Optional: open trip details from a card tap.

- **Success:** Search runs from a valid form; participant reaches a swipe deck, trip details, or the no-match state without stuck validation; they correctly describe swipe-right as requesting a ride and report the cost they saw (or honestly say no cost was visible).
- **Cost update (Area 4, optional):** If the moderator prepared a ride where the estimate changes after another rider joins, ask the participant to compare before and after. Otherwise only the on-card estimate in this task is required.

6.4 Scenario UT-2R: Recurring ride (reference only)

Not in scope. Documented for future runs. In this study, drivers chose a one-time ride in UT-3.

- **Driver (future):** Post Ride → recurring → set direction, departure, seats → publish.
- **Rider (future):** From My Rides on a recurring booking, use request-next when shown.

6.5 Scenario UT-3: Driver — post a ride (Area 3)

- **Start:** Driver trips list; open Post Ride (post-a-ride modal).
- **Prerequisite:** Home address on profile; moderator uses a seeded account or allows profile setup time.
- **Task:** Choose direction (to or from McMaster); keep repeat on **one-time**; set departure and seats; publish.
- **Success:** Publish completes (modal closes or trip appears in the list) without blocking errors, or the participant gets a clear error that the moderator logs as a defect.

6.6 Scenario UT-4: Accept, decline, or blocked match (Area 3; Area 4 if cost shown)

Moderator sets state so the participant has a pending match, request, or trip action.

- **Rider path (examples):** Confirm or cancel a request; open a matched trip; respond to prompts that accept or end the match. If the UI shows a cost or status change, note whether it is clear.
- **Driver path (examples):** Open pending requests for a trip; **accept** one rider or **reject** one when the script asks; confirm the UI reflects the choice (e.g. confirmed vs. rejected).
- **No match / blocked:** If the script uses a dead-end state, success is correctly describing the message and next step.
- **Success:** The participant reaches the intended state (accepted, declined, cancelled, or no match) or clearly explains why the screen blocked them; moderator notes any ambiguity.

6.7 Scenario UT-5: Profile update (Area 2)

- **Task:** Change a non-sensitive field (for example phone or display name) and save. If time allows, locate notification or payment-related settings and report whether they exist and where.
- **Success:** Update saves as expected, or failure is documented; settings located within the time threshold in Section 8 or participant explains why they stopped.

6.8 Scenario UT-6: Review after a trip (Area 5)

- **Prerequisite:** Moderator places the account on a completed or review-eligible trip (same build as production test data).
- **Task:** Open the review flow for that trip, set rating(s), and submit.
- **Success:** Submission completes with success feedback (or the participant describes a blocking issue for the log).

6.9 Scenario UT-7: Safety report (Area 5)

- **Prerequisite:** Active or completed trip context where Report is available (rider ride screen, driver drive screen, or trip details per moderator script).
- **Task:** Open Report (safety flow), complete the steps the moderator specifies for the test environment, and submit.
- **Success:** Report submits with confirmation, or the participant describes a clear blocker.

Table 2: Task scenario summary (UT-2R not run; quantitative outcomes in Section 9).

ID	Focus	Notes
UT-1	Verification and input errors	Areas 1
UT-2	Rider search, cost, pass/request	Areas 3, 4
UT-2R	Recurring ride	Not in scope
UT-3	Driver post ride	Area 3
UT-4	Match accept / decline / blocked	Areas 3, 4
UT-5	Profile update	Area 2
UT-6	Post-trip review	Area 5
UT-7	In-app report	Area 5

7 Session Questions and Survey Instruments

7.1 Pre-session Background Questions

Q1: How often do you commute to or from McMaster (or campus)?

Q2: Do you usually drive, ride as a passenger, or use transit?

Q3: Have you used any rideshare or carpool apps before? Which?

Q4: What device are you using today?

7.2 Post-task Probing Questions

After each scenario (Section 6), ask as appropriate:

PT1: What did you think that screen was asking you to do?

PT2: Was anything surprising or unclear?

PT3: On a scale of 1 (very difficult) to 7 (very easy), how easy was that task? (Record numeric response.)

PT4: Did you feel you could trust the information the app showed (e.g., cost, other user, timing)?

PT5: After UT-2 or UT-4, if a match or trip showed a price: was that dollar amount clear and believable?

7.3 Post-session Interview Questions

PS1: Overall, how would you describe your experience using Hitchly today?

PS2: What would you change first if you could?

PS3: Would you use this for real commuting if it were available? Why or why not?

PS4: Any concerns about safety, privacy, or payments?

7.4 Standardized Questionnaires

- **System Usability Scale (SUS):** Administer the standard 10-item SUS immediately after tasks; compute the standard 0–100 score per participant. Full wording appears in Appendix 10.4.
- **Optional supplementary Likert blocks:** e.g., trust in campus verification and clarity of pricing. Use 5- or 7-point scales with anchors "Strongly disagree" / "Strongly agree"; include items in Appendix 10.4.

All numeric survey responses feed the quantitative analysis described in Section 8.

8 Metrics and Quantification

This section defines the measures used in Section 9.

8.1 Task Performance

- **Task success rate:** For each of UT-1 through UT-7, the share of participants who met the scenario success criteria. Report per task and overall. (UT-2R was not run.)
- **Time on task:** Time from moderator prompt to success or stop; report median, interquartile range, and notable outliers.
- **Assists:** Count light hints vs. substantive help; optional composite "error score" if the team defines weights up front.
- **Critical incidents:** Count problems by severity (blocks task / major / minor) using a rubric fixed before sessions.

8.2 Subjective Measures

- **SUS score:** Per participant and mean ± standard deviation across N.
- **Post-task ease (1–7):** Mean per task from PT3 (Section 7).
- **Optional Likert blocks:** Means per theme (e.g. trust, clarity).

8.3 Analysis

- Summarize numbers in tables; add charts if the course requires them.
- Tag themes in open-ended answers; link to screenshots or timestamps when recordings exist.
- Tie major issues to SRS FR/NFR or product areas when useful.

8.4 Thresholds (optional targets)

Record any targets set before testing (e.g. at least 80% success on UT-1, median SUS ≥ 68).

9 Results

9.1 Scope and limitations

Results summarize ten moderated sessions on two Android devices (Section 5). The sample is small ($N = 10$), all on Android, and focused on McMaster test accounts; findings describe this cohort and are not generalized to every rider or driver. **UT-2R** (recurring rides) was not run.

9.2 Participant summary

- **Sample:** $N = 10$ (P1–P10).
- **Role mix:** Six sessions emphasized **rider** tasks (UT-2); four emphasized **driver** tasks (UT-3). Everyone completed UT-1, UT-4, UT-6, UT-7, and UT-5 when the script applied; trip state was prepared for review and report tasks.
- **Environment:** Two Android phones; API on **Render**; database on **Neon**; accounts reset between participants when required.

9.3 Quantitative summary

Table 3: Task success ($N = 10$).

Task ID	Successes	Rate (%)
UT-1	8/10	80
UT-2	7/10	70
UT-3	8/10	80
UT-4	8/10	80
UT-5	9/10	90
UT-6	9/10	90
UT-7	8/10	80

Table 4: Time on task, seconds (median and IQR).

Task ID	**Median**	**IQR**
UT-1	102	78–135
UT-2	152	118–198
UT-3	88	71–112
UT-4	74	58–96
UT-5	42	33–58
UT-6	68	52–86
UT-7	61	48–79

Table 5: Moderator assists (light hint vs. substantive help), all tasks.

Assist type	**Count**	**Per participant (mean)**
Light hint	18	1.8
Substantive help	4	0.4

Table 6: SUS scores (0–100 scale), by participant.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
64	70	68	76	82	71	73	69	75	78

Mean = 72.6; standard deviation = 5.4; min = 64; max = 82.

Table 7: Post-task ease (1 = very difficult, 7 = very easy), mean by task.

After task	Mean ease
UT-1	5.7
UT-2	5.1
UT-3	5.6
UT-4	5.7
UT-5	6.0
UT-6	5.8
UT-7	5.5

9.4 Task notes (UT-1, UT-2, UT-3, UT-4)

UT-1: Eight of ten met the full scenario (sign-up, verification, and error-recovery prompts). Two needed extra help on Part B (weak-password or format messages) before they stated the rule clearly.
UT-2: Seven of ten finished without substantive help. Two stalled on campus drop-off labels; one left the empty-match state without retrying within the time box. Most located the dollar estimate on a card after a short scan; one overlooked it until nudged. Median time from opening find-a-ride to search: 48 s (IQR 38–62 s); median time from results to first intentional swipe: 41 s.
UT-3: Eight of ten published on the first try. Two needed a hint about the home-address callout. Median time from Post Ride to success: 72 s (IQR 59–94 s).
UT-4: Eight of ten reached the scripted accept, decline, or blocked outcome. Driver reject/accept labels were clear after the first exposure for most; one participant hesitated on confirmation wording.
UT-2R: Not evaluated.

9.5 Review and report (UT-6, UT-7)

UT-6: Nine of ten submitted a rating without substantive help. One needed the moderator to point to the review entry on trip details.
UT-7: Eight of ten completed the report flow. Two needed a hint to find Report from the active-ride screen; all who reached the form understood the

final submit step.

9.6 Qualitative observations

Themes: (1) Campus drop-off names are hard to map to real buildings. (2) To/from McMaster wording is fine after first use but slows people at first. (3) Swipe-to-match works once the hint text is seen. (4) Estimated cost on cards is trusted but easy to miss if participants focus on time and route. (5) Review and report are secondary actions; visibility from trip and ride screens matters.

Representative paraphrased comments:

- "There are a lot of drop-off names—I'm not sure which is closest to my class." (UT-2)
- "I thought 'to McMaster' meant something else until I tried it." (UT-2)
- "After the address warning, publishing was easy." (UT-3)
- "Swipe right to request made sense once I read the bottom line." (UT-2)
- "I didn't notice the price until you asked—it's small next to the time." (UT-2)
- "Report was buried; I expected it under profile." (UT-7)

Critical incidents: four **major** (UT-2 drop-off/modal navigation, one UT-7 discovery); ten **minor** (wording, back navigation, empty match state, review entry).

10 Discussion and Recommendations

10.1 Interpretation

Outcomes line up with the goals in Section 4. **UT-1** and **UT-5** stayed relatively easy; adding sign-up error checks slightly increased assists compared with a happy-path-only sign-up. **UT-2** remained the hardest task (lowest success, longest times), driven by drop-off choice and first-time direction wording. **UT-3** was quick once the address prerequisite was met. **UT-4** behaved as expected given prepared match state. **UT-6** and **UT-7** mostly succeeded but showed **discovery** issues: cost on cards is easy to skip, and Report is not where some participants first looked.

Trade-offs: Richer campus landmarks add choice load; swipe matching needs a clear first-run cue; secondary actions (cost, report) need stronger placement or labels without cluttering primary flows.

Limitations: $N = 10$; Android only; McMaster test accounts; **Render/ Neon** latency not stress-tested; **UT-2R** omitted; quotes are paraphrased session notes.

10.2 Recommendations

10.2.1 General (all task areas)

Issue	Evidence	Sev.	Proposed change	Status
Drop-off landmark discoverability	UT-2: 70%; ease 5.1; drop-off quotes	Maj.	Search/filter or map snippet for McMaster drop-offs; group by zone	Open
First-time direction semantics	UT-2: hesitation; route assists	Min.	Short inline tip for to/from McMaster	Open
Empty match recovery	UT-2: one participant did not retry	Min.	After empty state, suggest time or route tweak	Open
Address prerequisite visibility	UT-3: 2/10 hints; ease 5.6	Min.	Link callout to profile in one tap when empty	Open
Sign-up error clarity	UT-1: 8/10; Part B assists	Min.	Inline password rules and email format hint before submit	Open
Estimated cost salience	UT-2: cost overlooked until prompt	Min.	Stronger label or placement on match cards	Open
Report entry discovery	UT-7: 8/10; buried expectation	Min.	Repeat Report link on trip details; consider profile safety hub	Open
Review entry discovery	UT-6: 1/10 needed hint	Min.	Keep review CTA visible after trip completes	Open

Table 8: Cross-cutting usability recommendations.

10.2.2 Trip, search, and post-ride

Issue	Evidence	Sev.	Proposed change	Status
Search-to-results timing	UT-2 median 152 s total	Min.	Skeleton or clearer progress during search	Open
Publish confirmation	UT-3: 8/10 first try	Min.	Stronger success feedback; return to trip list	Open
Recurring rides	UT-2R not in scope	—	Cover UT-2R in a later test round	Planned

Table 9: Trip and search recommendations (UT-2, UT-3).

10.3 Traceability

Priorities map to SRS usability and clarity expectations (simple flows, clear trip and match actions, trustworthy pricing). Track fixes with issues/PRs (e.g. drop-off UX, empty-state copy, profile shortcut from post-ride callout, review/report entry points).

10.4 Post-study interface revisions

Following the usability sessions, the team carried out a general interface pass so the app would feel closer to familiar rideshare and carpool products participants already know. We reviewed patterns from widely used apps in that category and applied them where they fit Hitchly's campus scope. Colour and accent usage were aligned across rider, driver, and shared screens so the experience feels like one product rather than separate areas. Primary actions, toggles, and secondary options were given clearer labels and hierarchy, with less ambiguous wording on flows that had caused hesitation in testing (e.g. direction, matching, and trip follow-up). The aim was to reduce guesswork: users should see what each control does and move through posting, searching, and trip-related tasks without having to infer meaning from layout alone. These changes complement the targeted recommendations in the tables above rather than replace ongoing tuning of copy and deeper features.

Appendix — Survey Instruments and Consent

Consent Script (Outline)

- Purpose of the session; voluntary participation; right to stop.
- Recording (screen/audio) if used; storage and who may view data.
- No penalty for withdrawal; approximate duration.
- Contact for questions.

System Usability Scale (SUS)

For each item, use a 5-point scale from **Strongly disagree** (1) to **Strongly agree** (5). Score using the standard SUS formula (odd items: score = response -1; even items: score = $5-$ response; sum $\times 2.5$ for 0–100).

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Optional Likert Items (Trust and Clarity)

Use 7-point Likert scales (1 = Strongly disagree, 7 = Strongly agree).

1. I trust that other users on this app are affiliated with McMaster as claimed.
2. The app's pricing or cost information was clear.
3. I understood what would happen after I confirmed a ride or match.
4. I felt my personal information was handled appropriately for a student project prototype.

Session Log Template (Fields)

- Participant ID (anonymous); date; moderator; device/OS; build.
- Per task (UT-1 through UT-7 as applicable): start time, end time, success Y/N, assists, notes.
- SUS total; optional Likert means.